



```
In [1]: import numpy as np
```

```
In [2]: arr= np.arange(1,21)
```

```
In [3]: arr
```

```
Out[3]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,
              18, 19, 20])
```

```
In [7]: arr2=arr.reshape(4,5)
```

```
In [9]: arr2.ndim
```

```
Out[9]: 2
```

```
In [10]: arr2
```

```
Out[10]: array([[ 1,  2,  3,  4,  5],
                [ 6,  7,  8,  9, 10],
                [11, 12, 13, 14, 15],
                [16, 17, 18, 19, 20]])
```

```
In [16]: arr2[:,2]
```

```
Out[16]: array([ 3,  8, 13, 18])
```

```
In [18]: arr2.sum(axis=0)
```

```
Out[18]: array([34, 38, 42, 46, 50])
```

```
In [19]: arr2.sum(axis=1)
```

```
Out[19]: array([15, 40, 65, 90])
```

```
In [21]: arr2.mean(axis=0)
```

```
Out[21]: array([ 8.5,  9.5, 10.5, 11.5, 12.5])
```

```
In [23]: arr2.std()
```

```
Out[23]: np.float64(5.766281297335398)
```

```
In [28]: arr2.max(axis=0)
```

```
Out[28]: array([16, 17, 18, 19, 20])
```

```
In [27]: arr2.min(axis=1)
```

```
Out[27]: array([ 1,  6, 11, 16])
```

```
In [29]: arr1= np.array([[1,2], [2,3]])  
arr2= np.array([[3,4], [5,6]])
```

```
In [30]: arr1
```

```
Out[30]: array([[1, 2],  
               [2, 3]])
```

```
In [31]: arr2
```

```
Out[31]: array([[3, 4],  
               [5, 6]])
```

```
In [32]: arr1+arr2
```

```
Out[32]: array([[4, 6],  
               [7, 9]])
```

```
In [ ]: arr1+100  
# [100,100][100,100]
```

```
Out[ ]: array([[101, 102],  
              [102, 103]])
```

```
In [34]: arr2+200
```

```
Out[34]: array([[203, 204],  
               [205, 206]])
```

```
In [43]: arr3= np.array([[100,200]])
```

```
In [47]: arr3
```

```
#[[100,200],[100,200]]
```

```
Out[47]: array([[100, 200]])
```

```
In [45]: arr2
```

```
Out[45]: array([[3, 4],  
               [5, 6]])
```

```
In [46]: arr2+arr3
```

```
Out[46]: array([[103, 204],  
               [105, 206]])
```

```
In [49]: arr2
```

```
Out[49]: array([[3, 4],  
               [5, 6]])
```

```
In [50]: arr1
```

```
Out[50]: array([[1, 2],
               [2, 3]])
```

```
In [51]: np.dot(arr1, arr2)
```

```
Out[51]: array([[13, 16],
               [21, 26]])
```

```
In [52]: #  $m \times n$   $n \times m$ 
```

```
In [53]: arr1
```

```
Out[53]: array([[1, 2],
               [2, 3]])
```

```
In [54]: arr2
```

```
Out[54]: array([[3, 4],
               [5, 6]])
```

```
In [55]: np.hstack([arr1, arr2])
```

```
Out[55]: array([[1, 2, 3, 4],
               [2, 3, 5, 6]])
```

```
In [56]: np.vstack([arr1, arr2])
```

```
Out[56]: array([[1, 2],
               [2, 3],
               [3, 4],
               [5, 6]])
```

```
In [57]: arr2
```

```
Out[57]: array([[3, 4],
               [5, 6]])
```

```
In [61]: np.hsplit(arr2, indices_or_sections=2)
```

```
Out[61]: [array([[3],
               [5]]),
          array([[4],
               [6]])]
```

```
In [65]: arr6=arr.reshape(4,5)
```

```
In [66]: arr6
```

```
Out[66]: array([[ 1,  2,  3,  4,  5],
               [ 6,  7,  8,  9, 10],
               [11, 12, 13, 14, 15],
               [16, 17, 18, 19, 20]])
```

```
In [68]: # np.hsplit(arr6, indices_or_sections=2)
```

```
In [69]: arr2
```

```
Out[69]: array([[3, 4],  
               [5, 6]])
```

```
In [70]: np.vsplit(arr2, indices_or_sections=2)
```

```
Out[70]: [array([[3, 4]]), array([[5, 6]])]
```

```
In [71]: arr6
```

```
Out[71]: array([[ 1,  2,  3,  4,  5],  
               [ 6,  7,  8,  9, 10],  
               [11, 12, 13, 14, 15],  
               [16, 17, 18, 19, 20]])
```

```
In [72]: arr6 %2==0
```

```
Out[72]: array([[False,  True, False,  True, False],  
               [ True, False,  True, False,  True],  
               [False,  True, False,  True, False],  
               [ True, False,  True, False,  True]])
```

```
In [73]: arr6[arr6 %2==0]
```

```
Out[73]: array([ 2,  4,  6,  8, 10, 12, 14, 16, 18, 20])
```

```
In [74]: arr6
```

```
Out[74]: array([[ 1,  2,  3,  4,  5],  
               [ 6,  7,  8,  9, 10],  
               [11, 12, 13, 14, 15],  
               [16, 17, 18, 19, 20]])
```

```
In [ ]:
```